

B.TECH PROJECT SYNOPSIS

Project Title

AI-Based Smart Attendance System Using Face Recognition

1. Introduction

Attendance management is an essential task... This project develops an AI-based attendance system using face recognition.

2. Problem Statement

Manual attendance is time-consuming and prone to errors and proxy attendance.

3. Objectives

- Develop automated attendance
- Detect and recognize faces
- Store records
- Generate reports

4. Proposed Methodology

Image Capture → Face Detection → Preprocessing → Feature Extraction → Face Recognition → Attendance Generation → Database & Report

5. Hardware Requirements

Intel i5/i7, 8 GB RAM, Webcam, 256 GB Storage

6. Software Requirements

Python, OpenCV, TensorFlow/PyTorch, VS Code, SQLite/MySQL

7. Expected Outcomes

Automated attendance, improved accuracy, reduced effort.

8. Applications

Schools, Colleges, Offices, Laboratories

9. Conclusion

10. References

- IEEE papers
- OpenCV Documentation
- TensorFlow Documentation